

Lab Technical Report: Secure SSH & Access Control

Implementation of Device Hardening, VTY Encryption, and Source Control Policy

Author: Eng. Mohamed Ashraf **Target Device:** SW2 (Cisco Catalyst Switch) **Topology:** Multi-Subnet Router-on-a-Stick **Date:** July 24, 2026

1. Executive Summary & Architecture

This lab demonstrates the complete end-to-end configuration and verification of secure management access on a Cisco Switch (**SW2**). The implementation covers basic device initialization, local credential management, line security, RSA key pair generation, standard Access Control List (ACL) binding to Virtual Typewriter (VTY) lines, and verification testing from both authorized and unauthorized sources.

Network Addressing & Reference Matrix

Device	Interface / Role	IP Address / Subnet	Default Gateway	Security Role
PC1	FastEthernet0	192.168.1.1 /24	192.168.1.254 (R1)	Authorized Management Console
R1	G0/1 & G0/0	192.168.1.254 / 10.0.0.1 /30	N/A	Core Inter-VLAN Router
R2	G0/0 & G0/1	10.0.0.2 /30 & 192.168.2.254 /24	N/A	Remote Router (Unauthorized Source)
SW2	VLAN 1 SVI	192.168.2.253 /24	192.168.2.254 (R2)	Hardened Switch Target
Laptop1	Console Connection	N/A (Console Serial)	N/A	Out-of-Band Admin Terminal

2. Switch Initial Configuration & Management SVI

Initial out-of-band administration was established from **Laptop1** via console port to configure hostname, secret credentials, SVI IP interface, and the default gateway.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#enable secret ccna
SW2(config)#username momo secret ccna
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.2.253 255.255.255.0
SW2(config-if)#no shutdown

%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SW2(config-if)#exit
SW2(config)#ip default-gateway 192.168.2.254
SW2(config)#exit
```

3. Console & Line Security Hardening

To protect against physical console unauthorized access and inactivity exposure, local database authentication and executive timeouts were configured.

```
SW2#conf t
SW2(config)#line console 0
SW2(config-line)#login local
SW2(config-line)#exec-timeout 3 0
SW2(config-line)#end
SW2#copy running-config startup-config
```

4. SSH Remote Access & Explicit ACL Filtering

To enforce encrypted remote administration and restrict management entry points strictly to administrator workstation PC1 (192.168.1.1), an RSA 2048-bit key pair was generated, and a Standard Access List was applied to VTY lines 0-15.

```
SW2#conf t
SW2(config)#ip domain name momo.com
SW2(config)#crypto key generate rsa modulus 2048
The name for the keys will be: SW2.momo.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable... [OK]

*Feb 28 19:16:41.1: %SSH-5-ENABLED: SSH 1.99 has been enabled

/* Define Access Policy: Permit PC1 ONLY */
SW2(config)#access-list 1 permit host 192.168.1.1

/* Configure VTY Lines for Secure SSH and Access Filtering */
SW2(config)#line vty 0 15
SW2(config-line)#login local
SW2(config-line)#exec-timeout 3 0
SW2(config-line)#transport input ssh
SW2(config-line)#access-class 1 in
SW2(config-line)#end
```

5. Verification Results & Access Testing

Test Case A: Access Verification from PC1 (192.168.1.1) PASSED

Ping connectivity to SW2 SVI (192.168.2.253) was verified, followed by an SSH session initiation. Since PC1 matches access-list 1 permit host 192.168.1.1, connection was granted successfully.

```
C:\>ping 192.168.2.253

Pinging 192.168.2.253 with 32 bytes of data:
```

```
Request timed out.
Reply from 192.168.2.253: bytes=32 time<1ms TTL=253
Reply from 192.168.2.253: bytes=32 time<1ms TTL=253
Reply from 192.168.2.253: bytes=32 time<1ms TTL=253

Ping statistics for 192.168.2.253:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l momo 192.168.2.253
Password: *****

SW2>en
Password: *****
SW2#conf t
SW2(config)# [CONNECTED & AUTHENTICATED SUCCESSFULLY]
```

Result: Authorized access from PC1 is fully operational. Encryption key standard: SSH 1.99 / 2048-bit RSA.

Test Case B: Security Enforcement Verification from R2 (10.0.0.2 / 192.168.2.254) **BLOCKED (EXPECTED)**

R2 tested connectivity via ICMP ping to SW2 (192.168.2.253) to confirm routing path availability. Upon attempting an SSH connection, SW2 enforced access-class 1 in, immediately dropping non-matching source packets.

```
R2#ping 192.168.2.253
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.253, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

R2#ssh -l momo 192.168.2.253
% Connection refused by remote host
R2#
```

Security Verification: Access Class filtering functioning as designed. R2 is rejected with explicit connection refusal, preventing rogue management sessions across intermediate layer-3 hops.

6. Conclusion & Recommendations

- **Objective Achievement:** All requirements (Console Hardening, Local Auth, 2048-bit SSH, and PC1-Only VTY Access Control) were successfully configured and validated.
- **Best Practices Note:** Ensure standard ACL 1 is mirrored or saved to startup configuration. For production enterprise deployments, consider using Extended Access Lists to filter on specific destination management ports (TCP 22) and AAA/RADIUS dynamic authorization.